Department of Computer Science & Engineering
The University of Texas at Arlington

System Requirements Specification
CSE 4316: Senior Design I
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EGG UTA
(Encapsulated Gardening Grounds)

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1 Product Concept

This section describes the basic functionality, purpose, use, and intended audience for the Encapsulated Gardening Grounds product, or EGG. The EGG is a product that serves to assist in small-scale individual crop growth. Users of the EGG will be able to better grow and care of their own personal indoor and outdoor garden.

The EGG describes an encapsulated plot of cropland and an associated web application. The physical base is equipped with sensors and automated systems, such as an irrigation system. Users can interact with the associated web application to parse the sensors' data, receive crop maintenance advice, and command manual use of the EGG's automated features.

1.1 Purpose and Use

The functionality above summarizes the primary purpose of the EGG product: To enable users to better care for their own, individual crops. Users of the EGG will be more self-sustainable and independent. EGGs are intended to be used in numerous environments, with their sensors enabling varied advice to users. Users will receive data, advice, and control over their crop specific to their environment.

1.2 Intended Audience

The intended audience for the Encapsulated Gardening Grounds (EGG) product includes individuals and small-scale growers who seek a more efficient and automated approach to personal gardening. This encompasses hobbyist gardeners, urban dwellers with limited outdoor space, and individuals interested in self-sustainability through homegrown crops. Additionally, the EGG is beneficial for beginners who may lack experience in plant care, as well as experienced growers looking for a streamlined way to monitor and maintain their crops. The integration of sensors and an interactive application makes it particularly appealing to tech-savvy users who value data-driven insights and remote management.

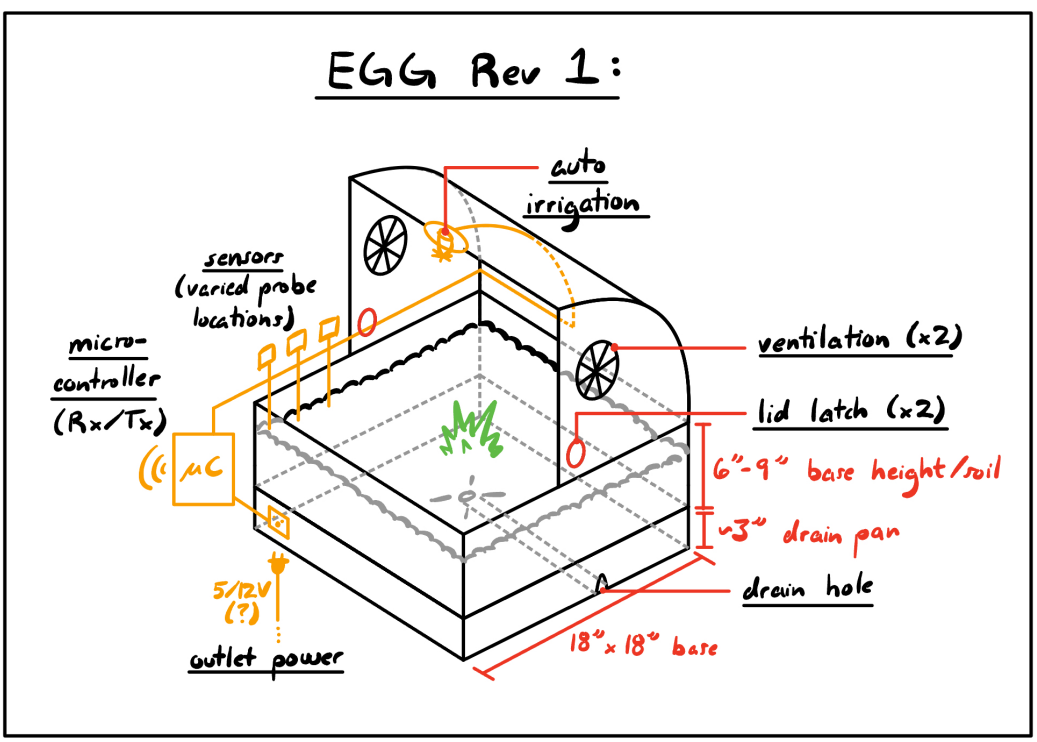

Figure 1: EGG Conceptual Drawing 1

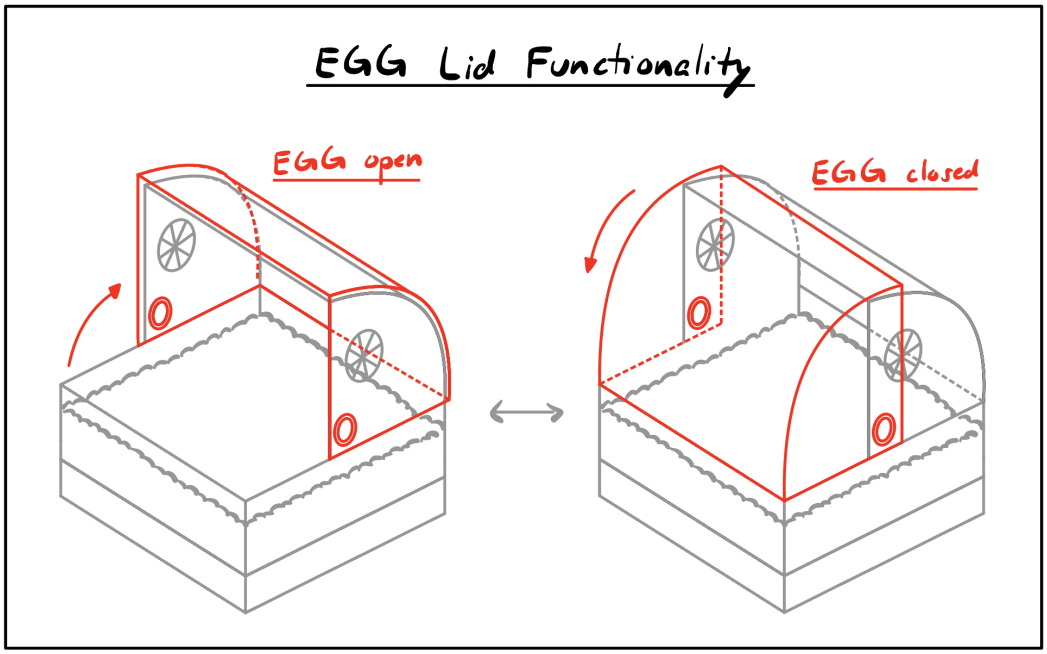

Figure 2: EGG Conceptual Drawing 2

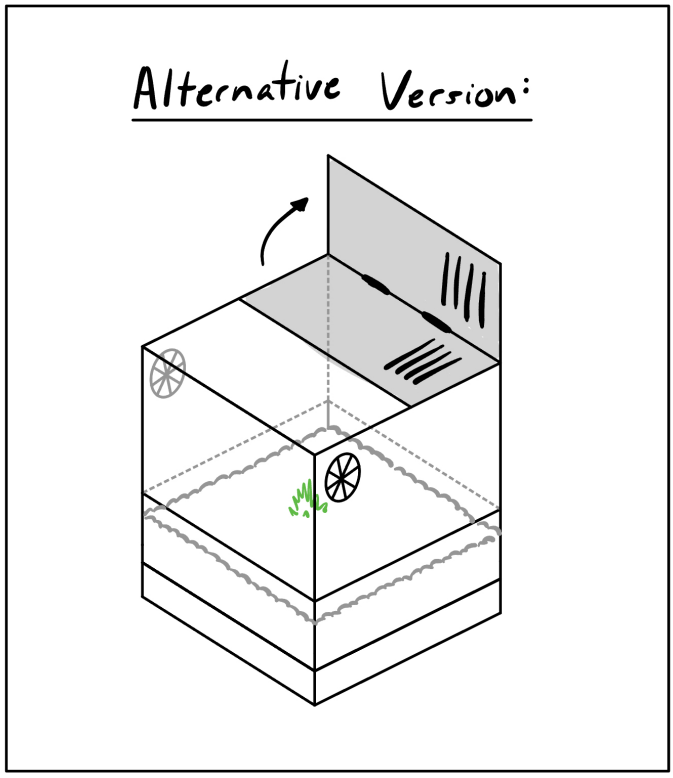

Figure 3: EGG Conceptual Drawing 3

2 Product Description

This section provides the reader with an overview of the Encapsulated Gardening Grounds, otherwise known as EGG. The primary operational aspects of the product, from the perspective of end users, maintainers and administrators, are defined here. The key features and functions found in the product, as well as critical user interactions and user interfaces are described in detail.

2.1 Features & Functions

The Encapsulated Gardening Grounds offer a simplified small-scale growing environment for individual use in indoor, urban, and exotic environments. With the goal of sustainable, individual, easy crop management, the EGG offers the following primary features and functions. Refer to the above, initial conceptual drawings of this product to better understand the functionality of these grounds:

- The project will involve the development of a controlled, enclosed environment optimized for plant growth.
- There will be an automated irrigation system that is controlled physically and through the associated desktop application
- Sensors in the physical EGG environment will be able to sample the crop's environment and parse the results through the application. Through this data, maintenance advice will be offered to the user, including suggestions to utilize the system's automated features. These automated features will be able to be used within the application.
- The enclosed environment holding the crop will be easy to interact with and allow for manual maintenance of the growing crop if needed.

2.2 External Inputs & Outputs

The outputs of the EGG are essential to its operation. Within its current scope, the physical EGG is intended to output varied, consistent samples of data on humidity, temperature, and more. This data is to be sent as an output and received as an input for the associated web application.
The web application will send out instructions for the automated features of the EGG, and the EGG will receive these instructions as input. This will allow for manual control of things such as the automated irrigation system as needed.

2.3 Product Interfaces

Users are expected to place their EGG in a suitable, sunlit environment and follow the directed advice and functions of the EGG when given. From the EGG's environment, data will be sent to the user's application, prompting them with advice to follow to best maintain the crop(s) growing in their EGG. This includes altering the basic care for the crop as needed: Providing more or less water, more or less sun exposure, etc. Some of these changes in care can also be accomplished through the user's input and interaction with the EGG's automated systems, such as the automatic irrigation.

3 Customer Requirements

The following requirements list the various features and functions for the Encapsulated Gardening Grounds' Customer Requirements. This includes the hardware and software that users will interact with for the product's primary functionality. Most importantly, these features constitute the data acquisition and hardware usage for the EGG. All of these features are essential to the goals of the product. This is the most current list of the requirements needed for the EGG product:

3.1 Desktop Application User Interface

3.1.1 Description

This requirement serves as a catch all need for the look and function of the application's front-end appearance. This interface will be interacted with by users to parse data from the EGG's sensors, view generated maintenance recommendations, adjust automated EGG features, and manually control those automated features. The goal of the user interface is to make it clear and easy for users to interact with the EGGs features with their limited knowledge. The palette of this user-friendly application will follow the color scheme outlined in other product materials. This includes the primary Orange (#FDA64E) and Cyan (#4FADB5) colors.

3.1.2 Source

Users of the EGG, including our UTA CSE sponsor, potential customers, and the project team's members must be able to utilize all of the EGG's features through this application interface. It is essential that these sources are to be satisfied with this requirement, in order to best utilize our Encapsulated Gardening Grounds.

3.1.3 Constraints

Depending on the scale of the application, some economic costs need to be considered. This includes any domain hosting that may be used. Other constraints include our desire to make the front-end user interface offer user accessibility. The application must be designed to be color-blind friendly. Additionally, the interface must be widely compatible. This means the interface must be able to be used across various browsers, offer wireless functionality, and work on mobile devices.

3.1.4 Standards

This requirement will institute the Website Content Accessibility Guidelines (WCAG) 2.1. This guideline outlines standards for allowing user applications to be user-friendly and accessible. [1]

3.1.5 Priority

This requirement has a CRITICAL priority. The user interface must be able to be used by users to get requisite information and features for the product.

3.2 Desktop Application Data Collection/Storage – Moisture Level

3.2.1 Description

This requirement describes the general back-end processing that goes into the Data Collection and Storage of the Moisture Levels within the Encapsulated Gardening Grounds (EGG). Data is collected from the micro-controller. The appearance and format of this data varies with the associated sensor that is gathering the data and being passed through from the micro-controller. In this way, the Moisture

Data must be received, parsed, and handled correctly. This data will be stored as needed in a format that is usable and visible to users through the application's user interface and front-end features.

3.2.2 Source

While not directly interacted with by end-users, this data is necessary for the Team to be able to parse and operate on. This data is essential to give a formatted, helpful version of the Moisture Levels for front-end viewing. In this way, the Team is dependent on this requirement being fulfilled.

3.2.3 Constraints

Requisite hardware will be utilized to best store and analyze this data. Economic costs arise from the implementation of this data collection and storage.

3.2.4 Standards

When handling user crop data, the ISO 27001 (Information Security Management) standard will be followed. This standard outlines general usage guidelines for user data collection and security. Data we be ensured to stay private under this standard. [2]

3.2.5 Priority

This requirement has a HIGH priority. Being able to read and utilize Moisture Level Data is an essential feature for the desktop application.

3.3 Desktop Application Data Collection/Storage – Nutrition Level

3.3.1 Description

This requirement describes the general back-end processing that goes into the Data Collection and Storage of the Nutrition Levels within the Encapsulated Gardening Grounds (EGG). Data is collected from the micro-controller. The appearance and format of this data may vary with the associated sensor that is gathering the data and being passed through from the micro-controller. In this way, the Nutrition Data must be received, parsed, and handled correctly. This data will be stored as needed in a format that is usable and visible to users through the application's user interface and front-end features.

3.3.2 Source

While not directly interacted with by end-users, this data is necessary for the Team to be able to parse and operate on. This data is essential to give a formatted, helpful version of the Nutrition Levels for front-end viewing. In this way, the Team is dependent on this requirement being fulfilled.

3.3.3 Constraints

Requisite hardware will be utilized to best store and analyze this data. Economic costs arise from the implementation of this data collection and storage.

3.3.4 Standards

When handling user crop data, the ISO 27001 (Information Security Management) standard will be followed. This standard outlines general usage guidelines for user data collection and security. Data we be ensured to stay private under this standard. [2]

3.3.5 Priority

This requirement has a HIGH priority. Being able to read and utilize Nutrition Level Data is an essential feature for the application.

3.4 Desktop Application Data Collection/Storage – Humidity Level

3.4.1 Description

This requirement describes the general back-end processing that goes into the Data Collection and Storage of the Humidity Levels within the Encapsulated Gardening Grounds (EGG). Data is collected from the micro-controller. The appearance and format of this data may vary with the associated sensor that is gathering the data and being passed through from the micro-controller. In this way, the Humidity Data must be received, parsed, and handled correctly. This data will be stored as needed in a format that is usable and visible to users through the desktop application's user interface and front-end features.

3.4.2 Source

While not directly interacted with by end-users, this data is necessary for the Team to be able to parse and operate on. This data is essential to give a formatted, helpful version of the Humidity Levels for front-end viewing. In this way, the Team is dependent on this requirement being fulfilled.

3.4.3 Constraints

Requisite hardware will be utilized to best store and analyze this data. Economic costs arise from the implementation of this data collection and storage.consideration.

3.4.4 Standards

When handling user crop data, the ISO 27001 (Information Security Management) standard will be followed. This standard outlines general usage guidelines for user data collection and security. Data we be ensured to stay private under this standard. [2]

3.4.5 Priority

This requirement has a HIGH priority. Being able to read and utilize Humidity Level Data is an essential feature for the application.

3.5 Automated Water Irrigation Adjustment

3.5.1 Description

This requirement lists a hardware feature/requirement for the Encapsulated Gardening Grounds. The EGG will implement water irrigation that is driven both manually (connected to a microprocessor over a wired and wireless connection) and automatically (on a programmed routine). The irrigation system will deliver water to both a sprinkler and drip irrigation output as needed. The sprinkler will spray water to provide hydration and increase the humidity of the EGG with an automatic or manually driven input from the user through the web application. This routine is determined with collected data from the moisture and humidity sensors once parsed through the desktop application.

3.5.2 Source

As a prominent feature of our product to be partly interacted with by users, this requirement is necessitated by both the team creating it, and any customer to use it. This includes our UTA CSE sponsors and the product's more general customers.

3.5.3 Constraints

The irrigation system is somewhat dependent on both social and economic factors. Socially, the hardware controls must be at least partially implemented in order to connect and use some of the various features related to this requirement's automation. This delay and dependency must be considered. Economically, the cost of the pumps, tubing, and spouts required for the irrigation methods must be quickly incurred when it comes to implementing this feature.

3.5.4 Standards

As a product that enhances the growth and production of food items, we find it necessary to follow the CEA's guidelines over Commodity Specific Food Safety, specifically for Controlled, Environment Agriculture. The Controlled Environment Agriculture (CEA) Alliance's advice towards such system is warranted and applicable to this product. In this way, this requirement's automation will follow these guidelines. [3]

3.5.5 Priority

This requirement has a HIGH priority. Being able to utilize the Automatic Irrigation System is an essential feature for the Encapsulated Gardening Grounds.

3.6 Automated Plant Nutrition Adjustment

3.6.1 Description

This requirement lists a hardware feature/requirement for the Encapsulated Gardening Grounds. The EGG will implement nutritional disbursement that is driven both manually (connected to a microprocessor over a wired and wireless connection) and automatically (on a programmed routine). The nutrition system will deliver fertilizer or nutritional solution(s) to the EGG as needed to promote crop growth. This routine is determined with collected data from the moisture and humidity sensors once parsed through the desktop application.

3.6.2 Source

As a prominent feature of our product to be partly interacted with by users, this requirement is necessitated by both the team creating it, and any customer to use it. This includes our UTA CSE sponsors and the product's more general customers.

3.6.3 Constraints

The nutritional system is somewhat dependent on both social and economic factors. Socially, the hardware controls must be at least partially implemented in order to connect and use some of the various features related to this requirement's automation. This delay and dependency must be considered. Economically, the cost of the pumps, tubing, and spouts required for the nutritional solutions disbursement must be quickly incurred when it comes to implementing this feature.

3.6.4 Standards

As a product that enhances the growth and production of food items, we find it necessary to follow the CEA's guidelines over Commodity Specific Food Safety, specifically for Controlled, Environment Agriculture. The Controlled Environment Agriculture (CEA) Alliance's advice towards such system is warranted and applicable to this product. In this way, this requirement's automation will follow these guidelines. [3]

3.6.5 Priority

This requirement has a HIGH priority. Being able to utilize the Automatic Nutrition System is an essential feature for the Encapsulated Gardening Grounds.

3.7 Automated Humidity Adjustment

3.7.1 Description

This requirement lists a hardware feature/requirement for the Encapsulated Gardening Grounds. The EGG will implement humidity adjustments that are driven both manually (connected to a microprocessor over a wired and wireless connection) and automatically (on a programmed routine). The humidity adjustment system will alter the humidity of the EGG as needed. The sprinkler will spray water as the primary way to increase the humidity of the EGG with an automatic or manually driven input from the user through the desktop application. This routine is determined with collected data from the moisture and humidity sensors once parsed through the web application.

3.7.2 Source

As a prominent feature of our product to be partly interacted with by users, this requirement is necessitated by both the team creating it, and any customer to use it. This includes our UTA CSE sponsors and the product's more general customers.

3.7.3 Constraints

The humidity adjustment system is somewhat dependent on both social and economic factors. Socially, the hardware controls must be at least partially implemented in order to connect and use some of the various features related to this requirement's automation. This delay and dependency must be considered. Economically, the cost of the pumps, tubing, and spouts required for the humidity adjustments must be quickly incurred when it comes to implementing this feature.

3.7.4 Standards

As a product that enhances the growth and production of food items, we find it necessary to follow the CEA's guidelines over Commodity Specific Food Safety, specifically for Controlled, Environment Agriculture. The Controlled Environment Agriculture (CEA) Alliance's advice towards such system is warranted and applicable to this product. In this way, this requirement's automation will follow these guidelines. [3]

3.7.5 Priority

This requirement has a HIGH priority. Being able to utilize the Automatic Humidity Adjustment System is an essential feature for the Encapsulated Gardening Grounds.

3.8 Desktop Application Hardware Controls

3.8.1 Description

This requirement is derived from a need from the Encapsulated Gardening Grounds' automated features. The Web Application must be able to control the various automated features of the EGG both manually (connected to a microprocessor over a wired and wireless connection) and automatically (on a programmed routine). The desktop application must able to connect to the EGG's micro-controllers

as needed to achieve this functionality. The user will be able to control the EGG's automated features with the desktop application through the application's user interface.

3.8.2 Source

Users of the EGG, including our UTA CSE sponsor, potential customers, and the project team's members must be able to utilize all of the EGG's features through this desktop application interface and its hardware controls. It is essential that these sources are to be satisfied with this requirement, in order to best utilize our Encapsulated Gardening Grounds.

3.8.3 Constraints

In order to best create a routine for which the Web Application will drive the automated features, data from the EGG's sensors are required. Additionally, the ability to control the automated features via the EGG's micro-controller and connected peripherals are required. These prerequisites must be implemented in order to achieve the best functionality out of the Encapsulated Gardening Grounds.

3.8.4 Priority

This requirement has a HIGH priority. Being able to utilize the EGG's Automatic Systems is an essential feature for the desktop application.

3.9 Desktop Application Plant Selection and Maintenance Advice Adjustment

3.9.1 Description

This requirement describes the ability to alter the chosen plant that a user might plant within the Encapsulated Gardening Grounds. A user will be able to interact with the EGG's associated Web Application to alter the chosen plant that is to be grown within the EGG. Derived from a list of implemented ideal conditions for the various plants supported by the EGG's Web Application, a user can pick the plant or crop that will be planted in the EGG. This choice will alter the desktop application's advice and maintenance routines to best suit the environment that the user's chosen plant will grow in. This includes altering the ideal humidity and moisture levels of the EGG's held plant, which will alter the automatic routines and given advice to the user through the associated desktop application.

3.9.2 Source

Users of the EGG, including our UTA CSE sponsor, potential customers, and the project team's members must be able to utilize all of the EGG's features through this Web Application interface and its hardware controls. It is essential that these sources are to be satisfied with this requirement, in order to best utilize our Encapsulated Gardening Grounds.

3.9.3 Priority

This requirement has a LOW priority. Being able to utilize the EGG's Automatic Systems is an additional feature for the Web Application.

3.10 Secondary EGG

3.10.1 Description

This requirement describes the desire for the team to recreate a secondary revision of the Encapsulated Gardening Grounds. Our ability to create a second revision of the EGG will allow us to make modifications and improvements to our established design and prove our ability to make a reproducible product.

A second revision will also enable the team to nail down documentation that is to be given to the user regarding the assembly and usage of the EGG, further proving the EGG as a viable, reproducible product.

3.10.2 Source

This requirement is a personal goal of the team in order to stomp out any perceived errors and create a more refined product to showcase to users, sponsors, and future peoples of interest.

3.10.3 Constraints

The most pertinent constraint to this requirement relates to our economics and time constraints. If we are to have requisite time and funds remaining to produce a second, revised version of the Encapsulated Gardening Ground, this stretch goal can be accomplished.

3.10.4 Priority

This requirement has a FUTURE priority. Being able to utilize the EGG is an additional goal that can enable revisions to the product's design, but is non-essential.

4 Material Requirements

This section provides requirements that are pertinent to the long-term functionality and usability of the device. Since the EGG may be used in harsher outdoor environments, these materials must be durable and resistant to the elements. Chemical safety pertaining to food and water safety, as well as electrical and electronics safety are also considered.

4.1 Frame and Base

4.1.1 Description

This requirement pertains to the frame and base of the device, which will contain water, plants, and soil during use. This container must be able to hold these materials for long periods without leaking, preventing damage to occur to sensitive electronics and messes from being made in the operating environment.

4.1.2 Source

Baseline requirement decided by the EGG Team.

4.1.3 Constraints

The frame and base of the device must be able to withstand long periods of use without leaks. This means the use of materials that are waterproof, weather-resistant, and rot-resistant. For purposes of growing produce, food-safe materials must be used to prevent the leakage of chemicals and plastics into the produce or water system.

4.1.4 Standards

As a product that enhances the growth and production of food items, we find it necessary to follow the CEA's guidelines over Commodity Specific Food Safety, specifically for Controlled, Environment Agriculture. The Controlled Environment Agriculture (CEA) Alliance's advice towards such system is warranted and applicable to this product. In this way, this requirement's automation will follow these guidelines[3].

4.1.5 Priority

This requirement has a HIGH priority. It is a base component that is crucial to the EGG's long-term use and durability of the EGG, and allows the user to interact with the functionality of the device. It also protects produce that may be grown and people who may consume it.

4.2 Encasement Cover

4.2.1 Description

This requirement pertains to the encasement material that will house a controlled environment for the crop. This encasement will allow the user access to view and handle the internal parts and crop of the device.

4.2.2 Source

Baseline requirement decided by the EGG Team.

4.2.3 Constraints

The encasement must not or negligibly affect the automatically controlled environment (i.e. conduct and introduce heat from the surrounding environment when undesirable. The encasement must be heat-resistant such that the material does not warp or discolor over time. The material must not be able to leak chemicals and plastics into the produce or water system. This material will ideally insulate the controlled environment while still allowing the user to see the EGG's contents.

4.2.4 Standards

As a product that enhances the growth and production of food items, we find it necessary to follow the CEA's guidelines over Commodity Specific Food Safety, specifically for Controlled, Environment Agriculture. The Controlled Environment Agriculture (CEA) Alliance's advice towards such system is warranted and applicable to this product. In this way, this requirement's automation will follow these guidelines[3].

4.2.5 Priority

This requirement has a HIGH priority. It is a base component that is crucial to the EGG's long-term use and durability of the EGG, and allows the user to interact with the functionality of the device. It also protects produce that may be grown and people who may consume it.

4.3 Sealant Materials

This requirement assumes the need for water sealant materials for water containment and plumbing. These materials may be used for the base or encasement, the irrigation system, or the recycling water supply. These could include lining materials, putties, pastes, finishes, or other materials.

4.3.1 Source

Baseline requirement decided by the EGG Team.

4.3.2 Constraints

Since the irrigation system will ideally recycle water, these materials must not pollute or leak chemicals and plastics into the water supply. The materials must also minimally affect the effectiveness of the controlled environment. It will also be considered that these materials should be replaceable, repairable, or both.

4.3.3 Standards

As a product that enhances the growth and production of food items, we find it necessary to follow the CEA's guidelines over Commodity Specific Food Safety, specifically for Controlled, Environment Agriculture. The Controlled Environment Agriculture (CEA) Alliance's advice towards such system is warranted and applicable to this product. In this way, this requirement's automation will follow these guidelines[3].

4.3.4 Priority

This requirement has a HIGH priority. It is a base component that is crucial to the EGG's long-term use and durability. It also protects produce that may be grown and people who may consume it.

4.4 Water Supply and Irrigation

4.4.1 Description

This requirement pertains to the water supply and irrigation system of the EGG, along with the plumbing required for operation. The water supply will recycle its water and must be able to be replenished by the user. Some or all parts of this system and its plumbing can be accessible to the user for maintenance.

4.4.2 Source

Baseline requirement decided by the EGG Team.

4.4.3 Constraints

Since the irrigation system will ideally recycle water, these materials must not pollute or leak chemicals and plastics into the water supply. The materials must also minimally affect the effectiveness of the controlled environment. The parts used in the water system must also be resistant to corrosion and temperature change.

4.4.4 Standards

As a product that enhances the growth and production of food items, we find it necessary to follow the CEA's guidelines over Commodity Specific Food Safety, specifically for Controlled, Environment Agriculture. The Controlled Environment Agriculture (CEA) Alliance's advice towards such system is warranted and applicable to this product. In this way, this requirement's automation will follow these guidelines[3].

4.4.5 Priority

This requirement has a HIGH priority. It is a base component that is crucial to the EGG's long-term use and durability. It also protects produce that may be grown and people who may consume it.

4.5 Electronics and Wiring

4.5.1 Description

This requirement pertains to the electronics and wiring of the automation and monitoring systems. These components must be capable of long-term use. The electronics and wiring must be robust enough to reliably power all needed peripherals including micro controller(s), sensors, pumps, etc.

4.5.2 Source

Baseline requirement decided by the EGG Team.

4.5.3 Constraints

Any electrical equipment must be waterproof or isolated from moisture. This may require a separate encasement for controllers and other essential electronics, sensors, and monitoring equipment rated for wet environments, insulated wiring, durable and moisture-resistant fans, among other possible considerations.

4.5.4 Standards

In cases where we may need more specific guidance on water-resistant enclosures, we will use IEC 60529, which describes the extent to which an enclosure protects against outside particles like dust or water.

4.5.5 Priority

This requirement has a HIGH priority. It is a base component that is crucial to the EGG's long-term use and durability. Since this is an automated system, the hardware must be highly resistant to failure.

4.6 Software Package

4.6.1 Description

This requirement pertains to the EGG's software systems. The device must be able to wirelessly connect to a database.

4.6.2 Source

Baseline requirement decided by the EGG Team.

4.6.3 Constraints

The software must be able to run and store operations and data on an Arduino. Software and drivers must be uploaded to the Arduino. This requires the user to be able to connect the device to an external computer or phone via wired, wireless, or Bluetooth connection. In the event of a wired connection, any port should be water-resistant and ideally have a cover.

4.6.4 Standards

In cases where we may need more specific guidance on water-resistant enclosures, we will use IEC 60529, which describes the extent to which an enclosure protects against outside particles like dust or water.

4.6.5 Priority

This requirement has a HIGH priority. It is a base component that is crucial to the EGG's long-term use and durability. Since this is an automated system, the hardware and software systems must be highly resistant to failure.

5 Performance Requirements

The EGG is designed to self-regulate its environment without the supervision of the user. However, materials such as the soil, the plants, and the water are not included in the packaged product. These items are expected to be provided by the user according to preference. As such, the performance requirements of the EGG will pertain to the maintenance of its environment and its communication with the user through the desktop app.

5.1 Desktop[] Application Activation

5.1.1 Description

The desktop application needs to be able to load in under 5 seconds. This is to ensure that users can quickly access information and an overview of the EGG's sensor measurements. If the activation time is longer than 5 seconds, it could indicate incomplete data, a function that consumes unnecessary memory, or even a mishandled error.

5.1.2 Source

The source is mainly the potential customers of the EGG who benefit greatly from a responsive system. As development continues, the team will aim to reduce this time by building the code as efficiently as possible.

5.1.3 Constraints

The loading time may be greatly affected by whatever data we choose to display on the home screen. Ideally, only simple records will be shown at first, such as the name of the selected plant, how long since it was last watered, and how many days since it was planted. More details would require a search through the sensors' last collected readings, which may take time to locate and display from the database storing those outputs.

5.1.4 Standards

We will follow the HTML Living Standard for our desktop application design. [4]

5.1.5 Priority

The priority is LOW. It is more important for the application to be able to run without freezing or stopping; we will work towards this requirement for a more functional EGG experience.

5.2 Manual Control Operation

5.2.1 Description

The desktop app allows users to manually control the EGG's features such as humidity, nutrients, and hydration supplied by various mechanisms. These controls are done via buttons on the User Interface that provide input to the EGG's system. If the feature was previously set to automatic, then the manual button should cause the feature to cease immediately and relinquish control to the user. This process should take under 2 seconds, and notify the user of the change.

5.2.2 Source

This requirement has been decided by the EGG team to facilitate a complete product. Manual control is imperative to the success of the EGG's operation.

5.2.3 Constraints

This requirement may be subject to safety concerns over the amount of energy required for the various subsystems to operate. Pushing the output of the microprocessors for the sake of speedy activation could cause shortages.

5.2.4 Standards

We will follow the HTML Living Standard for our desktop application design. [4]

5.2.5 Priority

This priority is HIGH. Although the buttons should be responsive, ensuring that manual control is available is the baseline of this requirement. Speedy response will be worked towards during the design of the desktop app.

5.3 Water Usage

5.3.1 Description

The EGG contains a drain pan that is 18" x 18" and 3" in depth. This area doubles as a source of water for the irrigation system, and a drain for used water that makes its way through the soil. This used water is rich in nutrients, and can safely be reused so long as there are no contaminants collected from other surfaces or fertilizers. This requirement specifies that approximately 4 gallons of water can be held in the pan, which can sustain the EGG for anywhere between 3 to 6 days, depending on the type of plant. This requirement may be modified if the drain pan is given more volume (see Future Items).

5.3.2 Source

The source would be the EGG team and our UTA CSE department sponsors. Our goal is to make the EGG self-sustaining, so water usage must be closely monitored and conserved as much as possible.

5.3.3 Constraints

Our constraint would be the safety of the water, as detailed in the CEA standards, as well as the functionality of the drain pan. Making the pan too large could greatly increase the weight of the EGG, which is meant to be movable so customers may adjust its exposure to light and other factors.

5.3.4 Standards

We will follow the CEA Alliance Standard to ensure the safe reuse of water on edible crops. [3]

5.3.5 Priority

The priority is MEDIUM. Irrigation is a necessary feature that must be functional for the EGG. The efficient use of water is desirable for self-sustainability of the EGG, which involves maximizing the amount of time the EGG is able to operate without user intervention.

6 Safety Requirements

Requirements 6.1 and 6.2 were provided by the professor. The following requirements are specific to EGG and are focused on ensuring the device will not directly or indirectly harm the user or its developers.

6.1 Laboratory equipment lockout/tagout (LOTO) procedures

6.1.1 Description

Any fabrication equipment provided used in the development of the project shall be used in accordance with OSHA standard LOTO procedures. Locks and tags are installed on all equipment items that present use hazards, and ONLY the course instructor or designated teaching assistants may remove a lock. All locks will be immediately replaced once the equipment is no longer in use.

6.1.2 Source

CSE Senior Design laboratory policy

6.1.3 Constraints

Equipment usage, due to lock removal policies, will be limited to availability of the course instructor and designed teaching assistants.

6.1.4 Standards

Occupational Safety and Health Standards 1910.147 - The control of hazardous energy (lockout/tagout).

6.1.5 Priority

Critical

6.2 National Electric Code (NEC) wiring compliance

6.2.1 Description

Any electrical wiring must be completed in compliance with all requirements specified in the National Electric Code. This includes wire runs, insulation, grounding, enclosures, over-current protection, and all other specifications.

6.2.2 Source

CSE Senior Design laboratory policy

6.2.3 Constraints

High voltage power sources, as defined in NFPA 70, will be avoided as much as possible in order to minimize potential hazards.

6.2.4 Standards

NFPA 70

6.2.5 Priority

Critical

6.3 Ergonomic/portable design

6.3.1 Description

The general design of EGG will ensure that users can operate, maintain, and transport the device without risk of harm to themselves or the EGG itself. Some aspects of this requirement are as follows.

- Smooth edges: Users should not be able to cut or pinch themselves while moving the EGG or its cover. Ideally, the EGG will have a recommended method of carry designed in.
- Robust construction: Users should be able to reasonably operate or move the EGG without the risk of dropping it or breaking components. Likewise, it should not leak water.
- Safe-to-clean surfaces: Users should be able to comfortably clean and sanitize surfaces without the risk of damaging components. In other words, anywhere a user may need to routinely clean should be easily accessible and waterproof.

6.3.2 Source

Baseline requirement decided by the EGG Team.

6.3.3 Constraints

Although the team will design the EGG with this requirement in mind, we may need to redesign and replace components if they cannot be reconciled somehow and continue to cause problems. This would include using a more suitable shape or material for the failing component, or possibly reinforcing it with some other material. Considering this is the first iteration of the EGG, this requirement will likely cost more time than budget.

6.3.4 Standards

The team likely won't be following any official standards in particular to enforce this requirement, but we will perform plenty of rigorous testing nonetheless.

6.3.5 Priority

This is a CRITICAL priority requirement. The EGG will not be practical or presentable if it is dangerous or carelessly brittle.

6.4 Separation of water and electronic components

6.4.1 Description

In order to prevent shocks and damage, the EGG will be designed such that vulnerable electronic components will not be exposed to water. The team will achieve this by using components already professionally designed and tested for water safety, and taking precautions for all other electronic components that have not.

6.4.2 Source

Baseline requirement decided by the EGG Team.

6.4.3 Constraints

The components that should already satisfy this requirement, such as sensors and water pumps, shouldn't be of much concern. However, components that do not will need to be physically separated from the enclosed space that deals with water, and precautions must be taken to ensure these separations will

not leak while also not overly hindering the greater design of the EGG. This will take additional design effort and likely cost more.

6.4.4 Standards

In cases where we may need more specific guidance on water resistant enclosures, we will use IEC 60529, which describes the extent to which an enclosure protects against outside particles like dust or water.

6.4.5 Priority

This requirement has CRITICAL priority, as it is vital to protect the team and users, not to mention that vulnerable components will likely break if this requirement isn't implemented.

6.5 Maintenance and dispersal of water

6.5.1 Description

As we are designing a product to cultivate food, we need to take the utmost care that the water used in the cultivation remains clean and free of contaminants throughout its usage. This would include the following prerequisites.

- All locations where water is stored fulfills this requirement.
- Water is dispersed in a safe and clean manner that doesn't encourage contamination.
- All water is easily able to be tested for contaminants, and any water-touching surfaces can be cleaned and sanitized.

6.5.2 Source

Baseline requirement decided by the EGG Team.

6.5.3 Constraints

Although we will be using suitable materials and recommended practices, the greatest challenge in testing that we have actually fulfilled this requirement is growing food to completion with no incidents, as there is no guarantee of success and we will have to start over and determine what went wrong each failure. This may incur severe time costs.

6.5.4 Standards

For this requirement, we will be following recommendations from the CEA Alliance organization over cleaning, sanitation, and testing. [3]

6.5.5 Priority

This requirement has CRITICAL priority because we cannot allow users to get food poisoning.

6.6 Power and battery

6.6.1 Description

As of now, EGG is set to be ran by a power outlet and/or supply, and should be both safe and efficient. Should a later iteration implements a mode of operation using a rechargeable battery, then that battery should still follow the same standard established in the previous version.

6.6.2 Source

Baseline requirement decided by the EGG Team.

6.6.3 Constraints

Fulfilling this requirement relies on us ensuring the equipment and components we use are trusted and reliable. This may mean increased costs for well-tested equipment, but for the most part it should not pose a problem.

6.6.4 Standards

We will use reliable equipment that follows the energy efficiency standards set by the US Department of Energy. If we implement a rechargeable battery, the Team will follow ANSI C18.

6.6.5 Priority

This is a CRITICAL priority requirement, as power failures could severely impact the functionality of the EGG's components.

6.7 Safety-specific instructions

6.7.1 Description

As EGG is a tool for our users, instructions and information pertaining to safe operation should be compiled and provided to them. This should include advising on cleaning and maintenance, and what to look for with tainted food.

6.7.2 Source

Baseline requirement decided by the EGG Team.

6.7.3 Constraints

The information we provide should be comprehensive, yet easy to understand and not overbearing.

6.7.4 Standards

To ensure the Team provides accurate information, we will look to the CEA Alliance's recommendations as well as our own experience. [3]

6.7.5 Priority

This is a HIGH priority requirement. Although not necessarily needed for a learned user who generally knows what they are doing, it would be highly beneficial to users to learn about any specific quirks the EGG has, or to users who don't have much experience with what responsibilities EGG asks of them.

7 Maintenance & Support Requirements

The Encapsulated-Gardening Grounds (EGG) and its associated web application will require ongoing maintenance and support to ensure reliable operation for users. This includes documentation, troubleshooting resources, and accessibility of support materials to aid users in operating and maintaining the system. Proper guides and manuals will be provided to assist with usage, troubleshooting, and potential error resolution.

7.1 USER GUIDES AND DOCUMENTATION

7.1.1 Description

Comprehensive written guides for the use of the EGG and its associated web application will be provided. These guides will cover system setup, usage, troubleshooting steps, and maintenance procedures. The documentation will be accessible via the web application, and physical copies may also be distributed as needed.

7.1.2 Source

Baseline requirement decided by the EGG Team.

7.1.3 Constraints

- The documentation must be regularly updated to reflect software updates and hardware changes.
- The web-based guide must remain accessible to users even when offline functionality is limited.
- Printed versions may require periodic reprints if significant changes occur.

7.1.4 Standards

IEEE Standard for Software User Documentation

7.1.5 Priority

High - Essential for user experience, product adoption, and long-term support.

7.2 SOFTWARE MAINTENANCE

7.2.1 Description

Ongoing software maintenance will be required to ensure the EGG system operates reliably. This includes:

- Keeping the mobile/web application up to date with security patches, feature enhancements, and bug fixes.
- Updating the micro controller firmware to improve system performance, fix issues, and support new features.
- Ensuring analytical systems, such as sensor data processing, function correctly and efficiently.

7.2.2 Source

Baseline requirement decided by the EGG Team.

7.2.3 Constraints

- The system must remain operational during updates.
- Compatibility with existing hardware and software must be maintained during updates.
- Regular monitoring of analytical systems is required to prevent inaccuracies in plant care recommendations.

7.2.4 Standards

ISO/IEC/IEEE 12207 : Systems and software engineering - Software life cycle processes. [5]

7.2.5 Priority

High - Critical for system functionality, user experience, and long-term reliability.

7.3 TECHNICAL KNOWLEDGE & HARDWARE FAMILIARITY

7.3.1 Description

A deep understanding of the EGG system's hardware, design, and internal components is required for effective maintenance and support. Team members responsible for maintaining the system must be trained to handle hardware repairs, component replacements, and system optimizations to ensure reliable operation.

7.3.2 Source

Baseline requirement decided by the EGG Team.

7.3.3 Constraints

- Team members must go under regular training to keep up with the changed to the system.
- Proper documentation of internal system architecture and troubleshooting procedures must be maintained and regularly updated.
- Availability of replacement components may impact the ability to quickly address hardware failures.

7.3.4 Standards

IEEE Standard for Software User Documentation

7.3.5 Priority

High - Essential for system longevity, reliability, and effective troubleshooting.

8 Future Items

The following requirements in this section discuss the above requirements that have a very low priority for their completion, symbolizing their status as FUTURE stretch goals. These are requirements that would be good to implement, but are in no way essential to the project's primary goals:

8.1 Secondary EGG

8.1.1 Description

This requirement describes the desire for the team to recreate a secondary revision of the Encapsulated Gardening Grounds. Our ability to create a second revision of the EGG will allow us to make modifications and improvements to our established design and prove our ability to make a reproducible product. A second revision will also enable the team to nail down documentation that is to be given to the user regarding the assembly and usage of the EGG, further proving the EGG as a viable, reproducible product.

8.1.2 Source

This requirement is a personal goal of the team in order to stomp out any perceived errors and create a more refined product to showcase to users, sponsors, and future peoples of interest.

8.1.3 Constraints

The most pertinent constraint to this requirement relates to our economics and time constraints. If we are to have requisite time and funds remaining to produce a second, revised version of the Encapsulated Gardening Ground, this stretch goal can be accomplished.

8.1.4 Priority

This requirement has a FUTURE priority. Being able to utilize the EGG is an additional goal that can enable revisions to the product's design, but is non-essential.

References

[1] W. W. W. Consortium, "Web content accessibility guidelines (wcag) 2.1," *W3C Standards*, 2024.

[2] I. O. for Standardization and the International Electrotechnical Commission, "Iso/iec 27001:2022 information security, cybersecurity and privacy protection â information security management systems ârequirements," *ISO/IEC Standards*, 2022.

[3] C. E. A. Alliance, "Commodity specific food safety guidelines for controlled environment agriculture," *CEA Standards*, 2023.

[4] WHATWG, "HTML living Standard," https://html.spec.whatwg.org/multipage/, 2025, [Accessed 28-03-2025].

[5] IEC12207, "Iso/iec/ieee international standard 12207-2017 - systems and software engineering – software life cycle processes," *IEEE Standards Association*, 2025.